

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations
 May 2025

IS4001-1 – INFORMATION AND NETWORK SECURITY
 (For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Techniques used for deciphering a message without any knowledge of the enciphering details is _____.
 A) blind deciphering
 B) Steganography
 C) Cryptanalysis
 D) Transposition
- DES exhibits the classic _____ block cipher structure, which consists of a number of identical rounds of processing.
 A) Feistel
 B) SAC
 C) Shannon
 D) Rendell
- What does the term "diffusion" refer to in cryptography?
 A) Making the relationship between the ciphertext and the key complex
 B) Dissipating the statistical structure of the plaintext into the ciphertext
 C) Reversing the encryption process
 D) Using different keys for different parts of the message
- What type of cipher involves replacing each letter of the plaintext with another letter, according to a fixed shift?
 A) Monoalphabetic Cipher
 B) Playfair Cipher
 C) Caesar Cipher
 D) Hill Cipher
- _____ depend on the running time of the decryption algorithm.
 A) Mathematical attacks
 B) Chosen ciphertext attacks
 C) Timing attacks
 D) Brute-force attacks
- The key exchange protocol is vulnerable to a _____ attack because it does not authenticate the participants.
 A) one-way function
 B) time complexity
 C) man-in-the-middle
 D) chosen ciphertext
- We can adapt the _____ algorithm to compute the greatest common divisor of two polynomials.
 A) Euclidean
 B) Cyclic
 C) Abelian
 D) Associative
- The _____ of two numbers is the largest integer that divides both numbers.
 A) prime polynomial
 B) greatest common divisor
 C) lowest common divisor
 D) integral divisor
- What is the main advantage of Elliptic Curve Cryptography (ECC) compared to traditional public-key systems?
 A) It requires larger key sizes for the same level of security
 B) It provides stronger security with smaller key sizes
 C) It is based on more complex mathematical problems
 D) It requires less computation power to break the encryption

10. In decentralized key control, how are keys managed?
 - A) Through a central trusted authority
 - B) By individual users without a central control
 - C) Through a public key infrastructure
 - D) By generating new keys for every communication
11. Which protocol uses X.509 certificates for securing communications?
 - A) SSH
 - B) AES
 - C) SSL/TLS
 - D) PGP
12. In hierarchical key control, what is the role of the root authority?
 - A) It encrypts all messages exchanged between users
 - B) It validates the signatures of messages
 - C) It manages and distributes keys to subordinate authorities
 - D) It generates random keys for encryption
13. What is the purpose of two-factor authentication in remote user authentication?
 - A) To ensure the password is not easily guessed
 - B) To encrypt user data before transmission
 - C) To combine two methods of verifying a user's identity, such as something the user knows and something the user has
 - D) To manage user session keys
14. What is a key advantage of using Kerberos over traditional password-based authentication?
 - A) It requires fewer encryption keys
 - B) It allows for faster communication between users
 - C) It eliminates the need for password management at the server side
 - D) It is less susceptible to man-in-the-middle attacks
15. What is the role of a Digital Certificate in remote user authentication?
 - A) It encrypts the user's data during transmission
 - B) It manages user passwords securely
 - C) It is used to verify the identity of the certificate holder and establish trust
 - D) It grants access to all resources within an organization
16. Which of the following describes how Kerberos helps with remote user authentication?
 - A) It uses a simple password-based login
 - B) It encrypts the entire communication session
 - C) It employs a trusted third-party authentication server to issue tickets
 - D) It uses public-key encryption for authentication
17. Which cryptographic protocol is most commonly used in HTTPS for secure web communication?
 - A) SSH
 - B) TLS
 - C) IPsec
 - D) SFTP
18. What is the default port for SSH?
 - A) 21
 - B) 22
 - C) 23
 - D) 80
19. Which IPsec protocol provides both encryption and authentication?
 - A) AH
 - B) TLS
 - C) ESP
 - D) SSH
20. Which component of IPsec is responsible for negotiating security associations?
 - A) ESP
 - B) IKE
 - C) AH
 - D) TLS

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Apply the Hill cipher decryption method to decode the message "ZEBB" using the given key

$$K = \begin{bmatrix} 6 & 24 \\ 1 & 13 \end{bmatrix}$$

Additionally, analyze this example to demonstrate the vulnerability of the Hill cipher to a known plaintext attack.

8 L*3 1 2

- b) Demonstrate the Public-key cryptosystem which provides both authentication and secrecy. Also, describe applications of a Public-key cryptosystem. 8 L2 2 1
2. a) Using the Playfair cipher with the keyword "MONARCHY", construct the Playfair matrix and decrypt the ciphertext "GATLMZCLRQX". Show all the steps. Also, discuss its security. 8 L3 1 2
- b) Demonstrate that in the Feistel structure, the output of the first round of the decryption process corresponds to a 32-bit swap of the input to the sixteenth round of the encryption process. Additionally, analyze the security aspects of the Data Encryption Standard (DES) 8 L3 1 1
3. a) A sender wants to encrypt the message $M = 9$ using the RSA algorithm. The public key is generated using the prime numbers $p = 7$ and $q = 11$.
Perform the following steps:
- Choose a public exponent $e = 3$ and compute the private key d .
 - Encrypt the message $M = 9$.
 - Decrypt the ciphertext to retrieve the original message.
 - Describe three approaches to attack RSA mathematically
- Show all steps clearly. 8 L3 2 2
- b) Explain the overall structure of the Advanced Encryption Standard (AES) algorithm by describing the following:
- Block size and key size used in AES.
 - Number of rounds for different key sizes (AES-128, AES-192, AES-256).
 - Main transformation steps in each round:
 - Difference between the initial round, main rounds, and final round.
- Support your explanation with a labelled diagram of the AES encryption process. 8 L2 1 1

Unit – II

4. a) Alice and Bob decide to use the **ElGamal Cryptosystem** for secure communication. They publicly agree on:
- Prime number $p = 17$
 - Primitive root $g = 3$
- Alice selects a **private key** $x = 7$ and computes her public key y .
Bob wants to send the message $M = 5$ securely to Alice. He chooses a random number $k = 4$ for encryption.
Perform the following steps:
- Compute Alice's public key y .
 - Encrypt Bob's message $M = 5$, obtaining ciphertext $(C1, C2)$.
 - Decrypt the ciphertext to retrieve the original message M .
- Show all calculations clearly. 4 L3 2 2

- b) Alice and Bob want to securely exchange a key using the **Diffie-Hellman Key Exchange Algorithm**. They publicly agree on:

- Prime number $p = 29$
- Primitive root $g = 2$

Alice selects a **private key** $a = 5$, and Bob selects a **private key** $b = 7$.

Perform the following steps:

- i. Compute Alice's public key **A**.
- ii. Compute Bob's public key **B**.
- iii. Compute the shared secret key that both will use for secure communication.

Show all calculations clearly.

c)	Explain the process of public key distribution.	4	L3	2	2
		8	L2	3	1
5.	a) Explain the format of the X.509 certificate and explain all the fields.	8	L2	3	1
	b) Discuss the Technical and environmental shortcomings of Kerberos version 4 and how it is addressed in Kerberos version 5.	8	L2	4	1
6.	a) Provide an overview of Kerberos version 4 message exchanges.	8	L2	4	1
	b) With a diagram, illustrate two approaches in symmetric key distribution using asymmetric encryption.	8	L2	3	1

Unit – III

7.	a) Explain the TLS Record Protocol and TLS Handshake Protocol with their roles in secure communication. Support your answer with a diagram.	8	L2	5	1
	b) With a diagram, explain the SSH Transport Layer Protocol and its packet exchange process.	8	L2	5	1
8.	a) Explain the IPSec architecture and the ESP packet format. Support your explanation with a diagram.	8	L2	5	1
	b) Explain the functionality of Tunnel Mode and Transport Mode in IPsec and discuss the security services provided by IPsec.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
